

The Cost of a Steeper Link: Calibration, Stability, and Tie Mechanics in Large-Scale Pairwise Preference Ranking

Aviral Poddar*
aviral@example.org

Abstract

Large-scale preference leaderboards such as Chatbot Arena rank language models by maximum-likelihood Bradley–Terry (BT), inheriting the logistic link’s independence-of-irrelevant-alternatives assumption over a comparison pool that never stops growing. We test whether a flexible Thurstonian alternative—a lattice-based model that on pairwise data reduces to a steeper, natively tie-aware gap link—measurably improves refit stability, held-out calibration, or tie handling on 1.67 million Arena votes spanning 129 models and sixteen months. Every comparison was pre-registered: effect-size thresholds derived before fitting, synthetic detection gates passed before real data, verdict criteria fixed in advance. We find no demonstrable advantage in either direction. Refit stability under population growth is statistically indistinguishable (median paired rank-correlation difference $+0.00000$ across thirteen month-over-month refits); two structurally different tests agree that link shape is not discriminable at the ability gaps real matchups occupy (analytic ceilings of $0.23\text{--}0.31\times$ our minimum practical difference); and the calibration and tie-mechanism comparisons both end *inconclusive*, each dominated by a single-window episode rather than any systematic difference. Inside this null sit two structural findings: a steeper link’s single tie parameter entangles tie mass with decisive discrimination, whereas Davidson’s tie parameter is provably orthogonal to its logistic link; and under parameter uncertainty, steep links trade prediction-side overconfidence against an estimation-side shrinkage that once protected against a severe cold-start extrapolation error. One anomaly survives every attribution attempt: temporally localized excess dispersion under every link tested. Practically, cold-start coverage and nonstationary drift—not link choice—bind next-month predictability in deployments of this shape.

1 Introduction

Chatbot Arena [2] has become the de facto public scoreboard for large language models: users pose a prompt, two anonymous models respond, and the user votes for a winner or declares a tie. The platform converts these votes into a leaderboard by maximum-likelihood Bradley–Terry (BT) [1], fitting one latent ability per model under a logistic link. Our study window alone contains 1.8 million recorded battles across 129 models (1.67 million after the platform’s own deduplication filter), with new models entering the comparison pool continuously throughout.

BT buys its simplicity with a strong structural assumption. As a logit-family choice model it satisfies Luce’s independence of irrelevant alternatives [5]: the odds of model A beating model B are taken to be invariant to whatever else occupies the pool. Nothing about human preference data guarantees this, and a leaderboard whose pool grows by the month—forcing a global re-estimate over an ever-changing comparison graph—is precisely the setting where one might expect the assumption to bite: incumbent ratings could shift when entrants arrive, predicted win probabilities could miscalibrate for fresh models, and the tie outcome (roughly a fifth of quality votes) is not even part of BT’s native outcome space. The Thurstonian random-utility family [6] offers the classical alternative—each model draws a latent performance and the better draw wins—and Cotton’s lattice

*Author list and affiliations placeholder — to be finalized.

construction [3] makes that family computationally exact for arbitrary base densities, with ties arising natively as dead-heat mass rather than by parametric extension [4]. On pairwise data the lattice model reduces to a *gap link*: a probability curve in the ability difference, like BT’s logistic but with a different shape—steeper at a toss-up (slope 0.284–0.337 depending on its tie-band width, against the logistic’s 0.25) and carrying a native tie band. Whether that different shape is worth anything in a real deployment is an empirical question, and to our knowledge it had not been asked at this scale.

We asked it four ways, each pre-registered before any real-data number was computed: practical-significance thresholds derived from deployment-meaningful error sizes, detection gates validated on synthetic ground truth, and pattern-based verdict criteria fixed in advance. (RQ1) When new models enter, whose refit perturbs incumbent rankings less? (RQ2) Is either link shape more consistent with observed win rates across triples of models? (RQ3) Which link calibrates better on held-out votes, especially for recent entrants? (RQ4) Does a native dead-heat band fit the observed tie structure better than Davidson’s extension of BT?

The answer, delivered with unusual statistical redundancy, is that the link does not matter here. Refit stability is indistinguishable—across thirteen month-over-month refits the median paired rank-correlation difference is +0.00000, with the two methods exactly tied in five windows—while window-to-window variation in stability, governed by pool maturity, exceeds the between-method differences by roughly two orders of magnitude. Two structurally different analyses (an analytic calibration ceiling and an empirical triple-additivity test) independently agree that at the ability gaps real matchups occupy, no plausible member of the Thurstonian family is distinguishable from the logistic. The two remaining comparisons, held-out calibration and tie handling, both end *inconclusive* rather than equivalent—a distinction we maintain deliberately, because each verdict was dominated by a single anomalous window rather than by any systematic difference, and our pre-registered criteria refuse a call under exactly that pattern.

This paper’s contribution is that negative result, stated with the precision it deserves, together with what we found inside it. First, a structural asymmetry: a steep link’s single tie parameter entangles tie mass with decisive discrimination, while Davidson’s tie parameter is provably orthogonal to its logistic link—forcing the lattice band to absorb both—bad-scale tie mass measurably degrades its decisive accuracy. Second, a two-faced interaction between link steepness and parameter uncertainty: steep links make overconfident plugin predictions from noisy estimates, yet the same steepness acts as implicit shrinkage in estimation—in our data it once prevented a severe cold-start extrapolation error that cost BT dearly. Third, an open question: a temporally localized excess dispersion, present under every link we tested and attributable to none of the covariates we measured. And throughout, a deployment lesson that inverts the motivating question: what binds ranking quality in a system like Arena is not the link function at all, but cold-start coverage and nonstationary drift—both of which dwarf every between-method difference we measured. We believe the audit itself, and the pre-registration discipline that kept four successive verdicts honest, are as much the contribution as any single number.

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